Test 1 : Differentiation

EXERCISE 1

1. Sketch the graph of $y = e^x - 6$.

2. Hence discuss the value and geometric interpretation of :

a. $\lim_{x \to -\infty} (e^x - 6)$

b. $\lim_{x \to \infty} (e^x - 6)$

EXERCISE 2 Differentiate from first principles the function

$$f(x) = 3x^3 - x.$$

EXERCISE 3 Find $\frac{dy}{dx}$ if :

$$y = \frac{1 + 3x}{x^2 + 1}$$

EXERCISE 4 A potter sells x vases per month.
His monthly profit in Australian dollars (AUD) can be modelled by

$$P(x) = -\tfrac{1}{5}x^3 + 7x^2 - 120, \quad x \geq 0.$$

(a) Find the value of P if no vases are sold.

(b) Differentiate $P(x)$.

(c) Hence, find the number of vases that will maximize the profit.

EXERCISE 5

Let

$$f(x) = \frac{1}{x^3}, \quad g(x) = 1 - 2x \quad \text{and} \quad h(x) = (f \circ g)(x).$$

Find $h(x)$ and show that the gradient of $h(x)$ is always positive.

EXERCISE 6ONLY SL

The gradient function of

$$f(x) = (2x - b)^a$$

is

$$f'(x) = 24x^2 - 24x + 6.$$

Find the constants a and b.

EXERCISE 7ONLY HL

Find $\frac{dy}{dx}$ in each of the following :

(a) $4x^2 + y^2 = 5$

(b) $y^3 = x^2$